EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

This lecture analyzes simple non-recursive FIR structures (Moving Average, Comb filters) and provides a rigorous, multidimensional engineering comparison between FIR and IIR digital filters.

Pedagogical Strategy:

1. Derive the Moving Average (MA) filter frequency response and zero placement on the unit circle.
2. Formulate Comb Filters and Notch Filters built from recursive FIR structures.
3. Construct the comprehensive **FIR vs. IIR Comparative Matrix**: Phase linearity, stability, filter order/computational cost, hardware memory, sensitivity, and transient response.
4. Establish decision rules for selecting between FIR and IIR filters in industrial engineering systems.

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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Analyze** the frequency selectivity and pole-zero placement of Moving Average and Comb filters.
2. **Compare** FIR and IIR digital filters across all architectural and algorithmic metrics.
3. **Select** the optimal filter topology for specific application constraints (latency, phase linearity, computational budget).

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2. MATHEMATICAL FOUNDATIONS

2.1 Moving Average Filter

$$y[n] = \frac{1}{M}\sum_{k=0}^{M-1} x[n-k] \implies H(z) = \frac{1-z^{-M}}{M(1-z^{-1})}$$

$$H(e^{j\omega}) = \frac{1}{M}\frac{\sin(\omega M/2)}{\sin(\omega/2)}e^{-j\omega(M-1)/2}$$

- $M-1$ zeros are located on the unit circle at $z = e^{j2\pi k/M}$ for $k = 1, 2, \dots, M-1$.
- The pole at $z = 1$ exactly cancels the zero at $z = 1$.

2.2 Comprehensive FIR vs. IIR Comparison Matrix

Engineering Metric	FIR Filter	IIR Filter
Phase Response	Can guarantee **strictly linear phase** ($\tau_g = \text{const}$)	Inherently **non-linear phase** (phase distortion)
Stability	**Always BIBO stable** (All poles at origin $z = 0$)	Must be tested; poles can migrate outside $
Filter Order	**High order** required for sharp transition bands	**Low order** (5$\times$ to 10$\times$ lower than FIR)
Computational Cost	High MAC count per sample	Low MAC count (extremely efficient)
Hardware Realization	Non-recursive (no feedback loops); pipelineable	Recursive (feedback loops); accumulator bottlenecks
Quantization Effects	Low sensitivity; free from feedback limit cycles	High sensitivity; susceptible to limit cycles & overflow
Analog Emulation	Cannot easily emulate analog prototypes	Directly designed from analog Butterworth/Chebyshev

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3. WORKED NUMERICAL EXAMPLES

Example 25.1: Order Estimation Comparison (FIR vs. IIR)

Problem: Estimate the required filter order for an FIR filter (Kaiser) vs. an IIR filter (Butterworth) to meet: $\omega_p = 0.4\pi, ; \omega_s = 0.5\pi, ; A_p = 0.5 \text{ dB}, ; A_s = 50 \text{ dB}.$

Solution:

- **FIR (Kaiser Formula):** $\Delta f = \frac{0.5\pi - 0.4\pi}{2\pi} = 0.05.$
  $N_{\text{FIR}} \approx \frac{50 - 7.95}{14.36 \times 0.05} + 1 = \frac{42.05}{0.718} + 1 = 58.56 + 1 \approx \mathbf{60 \text{ taps}}.$
- **IIR (Butterworth Order Formula):** Prewarping / prototype ratio:
  $\frac{\Omega_s}{\Omega_p} = \frac{\tan(0.25\pi)}{\tan(0.2\pi)} = \frac{1.0}{0.7265} = 1.3764.$
  $N_{\text{IIR}} = \left\lceil \frac{\log_{10}[(10^5 - 1)/(10^{0.05} - 1)]}{2\log_{10}(1.3764)} \right\rceil = \left\lceil \frac{\log_{10}(100000/0.122)}{2 \times 0.1387} \right\rceil = \left\lceil \frac{5.9136}{0.2774} \right\rceil = \mathbf{22 \to 12\ (BLT)}.$
- **Takeaway:** The IIR filter requires ≈ 6 biquad sections (12 poles), needing significantly fewer MAC operations per sample than the 60-tap FIR filter.

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Provide a detailed comparative analysis between FIR and IIR digital filters covering Phase Linearity, Stability, Computational Complexity, and Hardware Implementation. *(10 Marks)* **(b)** Under what practical engineering scenarios is an FIR filter mandatory over an IIR filter? *(5 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):** Comprehensive comparison matrix covering all 7 parameters listed in Section 2.2 *(10 Marks)*
- **Part (b):** 1. Digital Communications (preventing Intersymbol Interference), 2. High-Fidelity Audio processing (preserving transient phase coherence), 3. Biomedical ECG/EEG diagnostic monitoring (preventing distortion of QRS peak morphology) *(5 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

# Order comparison
N_fir, beta = signal.kaiserord(50, 0.1)
N_iir, Wn = signal.buttord(0.4, 0.5, 0.5, 50)
print(f"Required FIR Order: {N_fir}")
print(f"Required IIR Order: {N_iir}")
```